Handling Missing Data in ETL

Q1. What are the most common reasons for missing data in ETL pipelines?

The most common reasons for missing data in ETL pipelines are incomplete data entry, extraction failures, integration of multiple data sources, transformation errors, system or file failures, and optional fields. Missing data reduces data quality and can lead to inaccurate reports and poor business decisions. Therefore, ETL pipelines should validate and handle missing values before loading data into the target system.

Q2. Why is blindly deleting rows with missing values considered a bad practice in ETL?

Blindly deleting rows with missing values is a bad practice because it can remove valuable information, reduce the accuracy of reports, introduce bias, and lead to incorrect business decisions. Instead, ETL pipelines should validate the data and use appropriate techniques such as imputation, default values, or flagging records before deciding whether to reject them.

Q3. Explain the difference between: Listwise deletion Column deletion Also mention one scenario where each is appropriate.

Listwise deletion removes entire rows that contain missing values, while column deletion removes an entire column if it has too many missing values. Listwise deletion is appropriate when only a few records are incomplete and complete records are required. Column deletion is appropriate when a column has a very high percentage of missing values and is not important for business analysis.

Q4. Why is median imputation preferred over mean imputation for skewed data such as income?

Median imputation is preferred over mean imputation for skewed data because the median is not affected by extreme values or outliers. In datasets like income, a few very high salaries can increase the mean significantly, making it unrepresentative. The median represents the middle value and provides a more accurate estimate for replacing missing values.

Q5. What is forward fill and in what type of dataset is it most useful?

Forward fill is a missing value imputation technique that replaces NULL values with the last available valid value. It is most useful for time-series or sequential datasets, such as stock prices, sensor data, weather records, and daily sales, where previous values remain valid until updated.

Q6. Why should flagging missing values be done before imputation in an ETL workflow?

Flagging missing values before imputation helps preserve information about which values were originally missing. It improves transparency, supports auditing, enables better analysis, and allows analysts or machine learning models to distinguish between original and imputed values. Once values are imputed without flagging, the original missing information is lost."

Q7. Consider a scenario where income is missing for many customers.

If income is missing for many customers, I would choose median imputation because income data is usually skewed and often contains outliers (very high or very low incomes).

The median is the middle value in the dataset and is not affected by extreme values, making it a more accurate and reliable estimate than the mean for replacing missing income values.

Q8. Listwise Deletion

Remove all rows where Region is missing.

DELETE FROM Customer_Data

WHERE Region = 'NaN';

In listwise deletion, we remove the entire row if it contains a missing value in a required column. In this dataset, only Customer_ID 105 has a missing Region, so that row is deleted. After deletion, the dataset contains 7 records, and 1 record is lost.

Q9. Imputation

Handle missing values in Monthly_Sales using: Forward Fill

Forward Fill replaces a missing value with the previous non-null value.

Before vs After

Customer_ID	Name	Monthly_Sales (Before)	Monthly_Sales (After)
101	Rahul Mehta	12000	12000
102	Anjali Rao	NaN	12000
103	Suresh Iyer	15000	15000

104	Neha Singh	NaN	15000
105	Amit Verma	18000	18000
106	Karan Shah	NaN	18000
107	Pooja Das	14000	14000
108	Riya Kapoor	16000	16000

Dataset After Forward Fill

Customer_ID	Name	City	Monthly_Sales	Income	Region
101	Rahul Mehta	Mumbai	12000	65000	West
102	Anjali Rao	Bengaluru	12000	NaN	South
103	Suresh Iyer	Chennai	15000	72000	South
104	Neha Singh	Delhi	15000	NaN	North
105	Amit Verma	Pune	18000	58000	NaN

106	Karan Shah	Ahmedabad	18000	61000	West
107	Pooja Das	Kolkata	14000	NaN	East
108	Riya Kapoor	Jaipur	16000	69000	North

Forward fill is suitable because:

- Monthly_Sales is a sequential/time-based value.
- It assumes the previous month's sales remain valid until a new value is available.
- It preserves all records instead of deleting them.
- It is commonly used in sales, financial, sensor, and time-series datasets.

Q10. Flagging Missing Data Create a flag column for missing Income.

SELECT COUNT(*) AS Missing_Income_Count

FROM Customer_Data

WHERE Income = 'NaN';

To flag missing income values, I use a CASE statement to create an Income_Missing_Flag column. Records with NULL income receive a flag of 1, and records with valid income receive 0. In the given dataset, there are 3 customers with missing income.